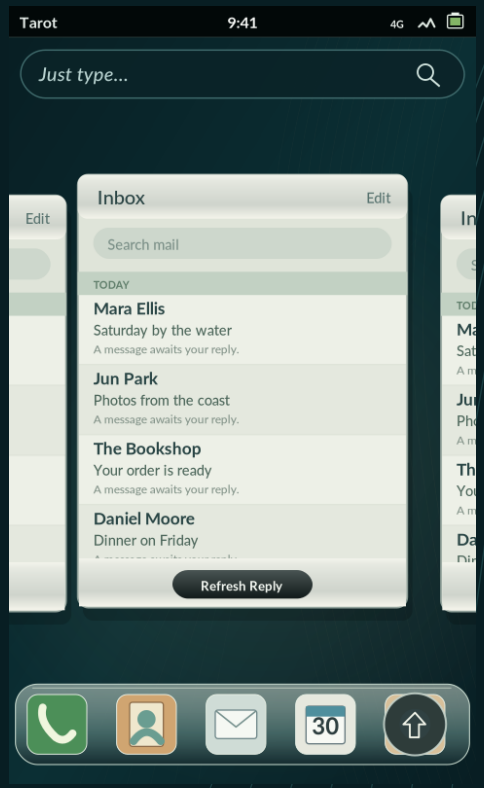
TAROT

PALM PRE 3 / DESIGN PACKET

A webOS-inspired phone shell on a modern Linux stack

Nocturne / native 480 × 800 / slide-out keyboard / hardware gestures



A separate edition for the original webOS hardware form.

Design thesis

Re-create the coherent phone interface; do not re-create the OS beneath it.

Fidelity before novelty

Cards, stacks, Quick Launch, Just Type and bottom notifications preserve the Pre 3 interaction grammar. No new app chrome is imposed on unrelated Wayland clients.

Own the shell, not the platform

Tarot owns composition and trusted UI. Nura/postmarketOS, system services and independent application toolkits remain responsible for their domains.

Hardware is a first-class input

The Pre 3 gesture sensor and physical keyboard remove two necessary compromises of the PinePhone Pro edition.

One system, multiple devices

The same UI components and activity model vary by hardware profile. Fx0 Gold remains Fx0-specific; Pre 3 defaults to Nocturne.

This is a proposed visual/interaction specification, not a running Pre 3 port or firmware capture.

The physical canvas

Documented hardware facts and requested porting assumption

DISPLAY	480 × 800 px; 3.58-inch WVGA; ~261 ppi
INPUT	Touchscreen + dedicated capacitive gesture area
TEXT	Slide-out QWERTY; Option, Shift, Sym, Enter
MEMORY	512 MB RAM; model-specific capacity may vary
PLATFORM	Nura/postmarketOS port assumed for design purposes
GRAPHICS	Adreno-class hardware; working DRM/GBM/GLES unverified

Hardware keys are outside the exported display pixels. Slider motion does not resize the LCD.

The slider and the sensor

Schematic shell study: keyboard slides below a fixed-size LCD.

TAROT / PRE 3

The gesture area lives below the display. The keyboard slides out, but LCD geometry does not change.



SLIDER CLOSED

480 × 800 LCD / external gestures



SLIDER OPEN / QWERTY

480 × 800 LCD / external gestures

Black-glass / charcoal hardware; schematic only, not dimensional CAD or original HP artwork.

The three phone profiles

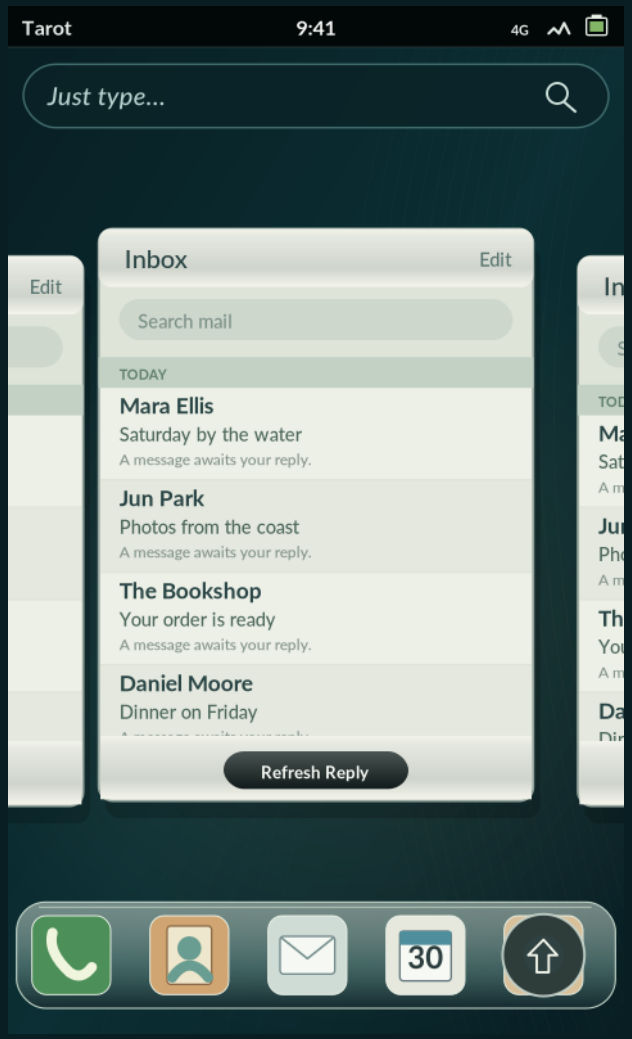
Shared design system, deliberately different input surfaces

ROLE	PINEPHONE PRO	LG Fx0	HP PRE 3
Output	720 × 1440	720 × 1280	480 × 800
Logical	360 × 720	360 × 640	360 × 600
UI scale	2.0	2.0	4/3
Gesture	On-screen strip	On-screen strip	Dedicated sensor
Extra input	Touchscreen	Home key	Slide keyboard
Primary IME	Software	Software	Hardware
Theme	Nocturne	Fx0 Gold	Nocturne

Pre 3 design removes software navigation reservation entirely. The device must report gesture input.

Anatomy of the interface

A logical canvas of 360 × 600; exports are native 480 × 800



Status 0-24

Black, centered clock, app/device corners

Just Type

Always visible in Card view; physical keys activate it

Card 53,130

Card is the same Wayland client, transformed

Quick Launch

Five slots, only four user-editable

Gesture area

Below LCD; zero screen pixels reserved

Typography and materials

Nocturne palette with restrained Palm-era surface treatment

Night / 081E23

Deep teal / 0B2F34

Warm paper / EAEDE3

Ink / 243537

Brass / D7BE89

Sea glass / 6FBCC0

Lato / everyday UI and status

TAROT / display identity

Prelude remains the historical comparison target, not an assumed licensed font.

Hardware gesture ownership

The physical gesture region is not a black software strip.

Center tap	Focused app ↔ Card view / selected card maximize
Short back swipe	Back only in cooperating app or declared bridge
Upward gesture	Overview; when already there, Launcher
Upward hold & drag	Quick Launch wave; release over favorite
Full swipe	Advanced switch-app behavior, when enabled
Fallback	Optional on-screen gesture control if driver is absent

Critical: the sensor needs an explicit Linux input-event contract; do not synthesize from app touch.

Keyboard-first interaction

The slide-out QWERTY changes the default state machine

Closed	Card view and Launcher retain Just Type field. No permanent on-screen IME.
Slider opens	Do not resize or rotate the panel. Keep current focus and surface coordinates stable.
Typing in home	Begin Just Type immediately; do not require a tap before typing.
Typing in app	Send physical key events to focused client; reserve global shortcuts deliberately.
Symbols / fallback	Honor Option, Shift and Sym. Software IME for accessibility or absent keys.
Slider closes	Keep edit state. Do not silently close a form or discard pending text.

Core visual system

Surface hierarchy should be recognizable before icons are inspected



Cards & stacks

Window content is continuous through maximize/minimize; stack members preserve independent identity and pending-close state.

Top corners

App actions are left, device services right. Menus are restrained, rounded, shadowed planes.

Bottom edge

Quick Launch rests in Card view; the Dashboard expands from the bottom while respecting content and touch ownership.

App integration boundaries

Semantic cards can be layered onto ordinary Wayland surfaces

Native clients

GTK, Qt, Rust and Waydroid (if viable) present ordinary Wayland windows.

Cooperating clients

Firefox and Thunderbird bridges may promote a tab or message to a distinct window.

Trusted shell

Tarot draws gesture feedback, Card view, status, app menus, notifications and lock.

System services

Nura/postmarketOS supplies audio, network, power, authentication and IPC.

Activity bridge

Optional app capabilities; no compulsory scriptable shell runtime.

Performance design for 512 MB

Avoid promising desktop-class behavior on a constrained 2011 handset

Render only what is needed

Keep visible native surfaces live. Favor direct texture reuse, bounded shadows, and no full-resolution history of hidden clients.

Initial cache policy

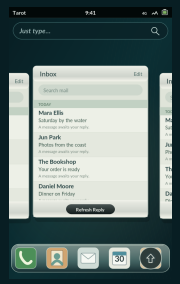
Three retained previews and 12 MiB owned cache are planning caps, not benchmarks. Measure actual resident GPU and process memory.

Waydroid is conditional

A Nura port assumption is not Android binder, GPU, memory or kernel compatibility. Keep Waydroid opt-in and disabled by default on Pre 3 pending evidence.

The screen atlas

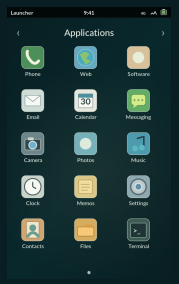
28 editable screen studies • 480 × 800 PNG and SVG variants



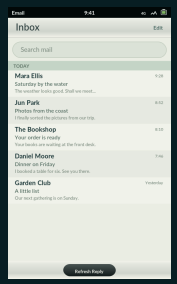
01 Card view



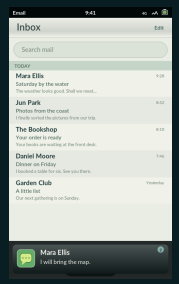
02 Card stack



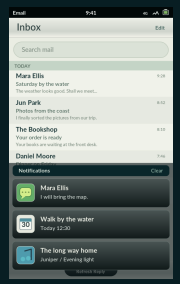
03 Launcher



04 Email



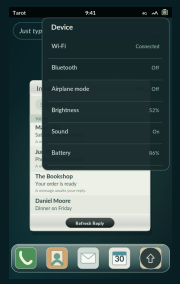
05 Notificati



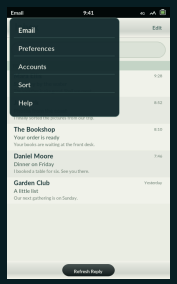
06 Dashboard



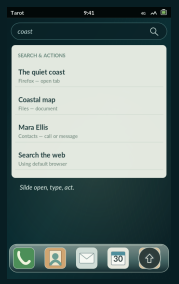
07 Quick Laun



08 Device men



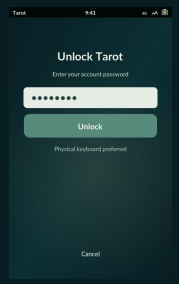
09 Applicatio



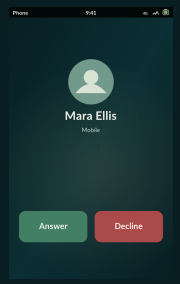
10 Just Type



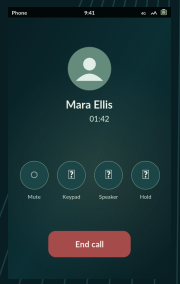
11 Lock scree



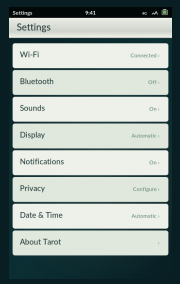
12 Authentica



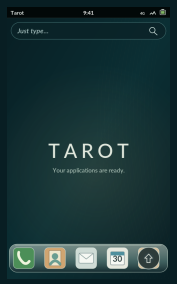
13 Incoming c



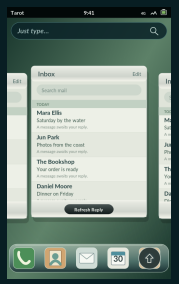
14 Active cal



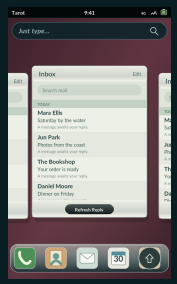
15 Settings



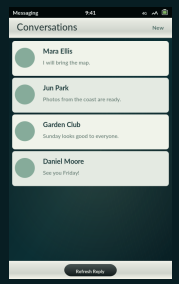
16 Empty card



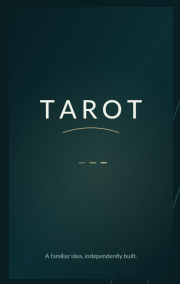
17 Heritage



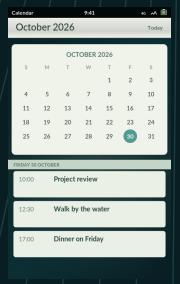
18 Vellum



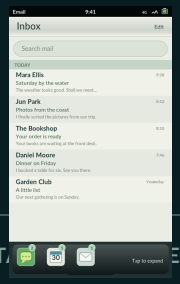
19 Messaging



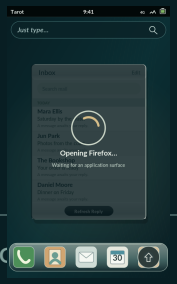
20 Boot



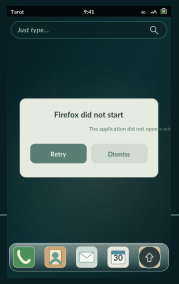
21 Calendar



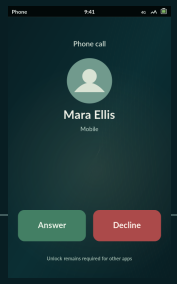
22 Notificati



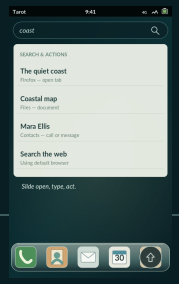
23 Launch pen



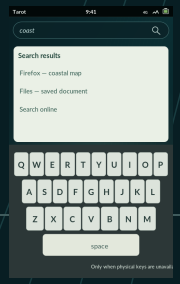
24 Launch fai



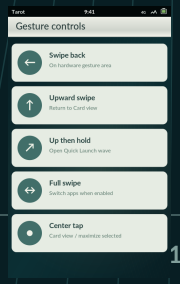
25 Locked inc



26 Slider ope



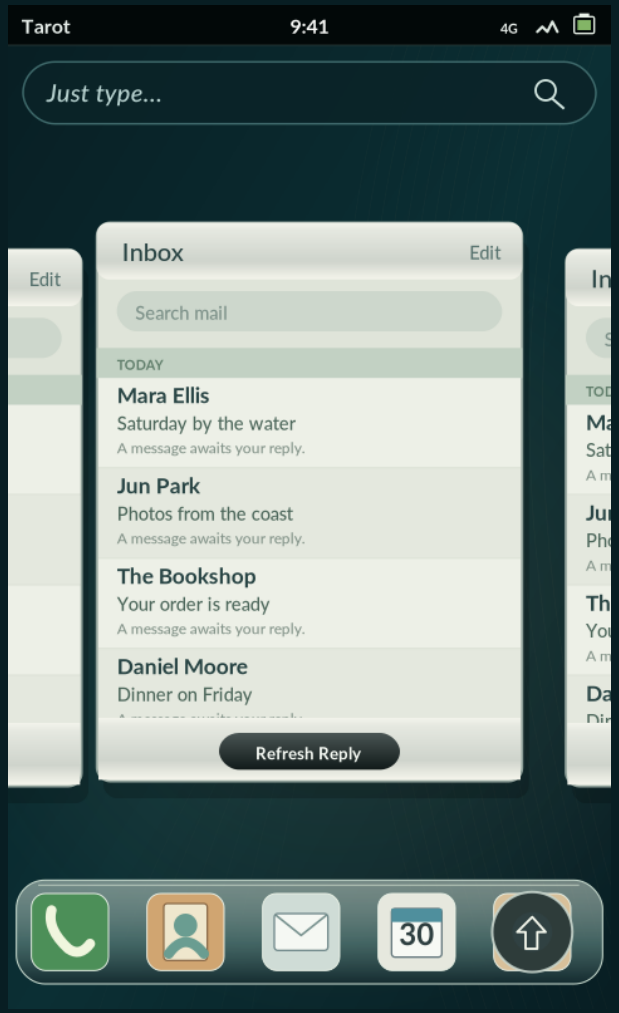
27 Keyboard f



28 Gesture ar

01–02 / screen studies

Editable SVGs preserve component geometry; PNGs verify native resolution.



Card view

One native client, another scale



Card stacks

Independent windows remain in a group

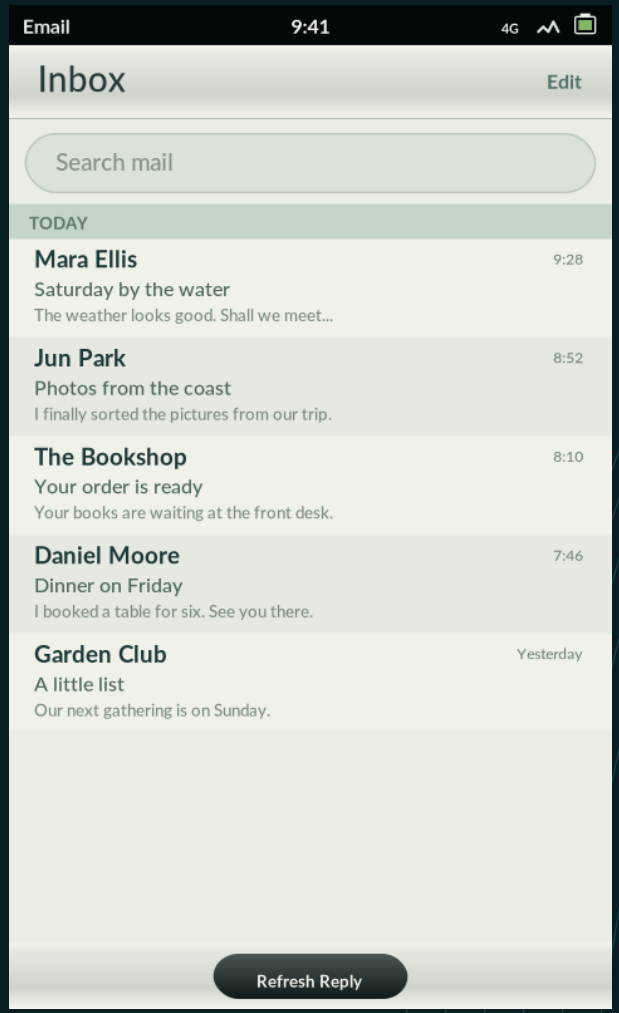
03–04 / screen studies

Editable SVGs preserve component geometry; PNGs verify native resolution.



Launcher

Three columns and named pages

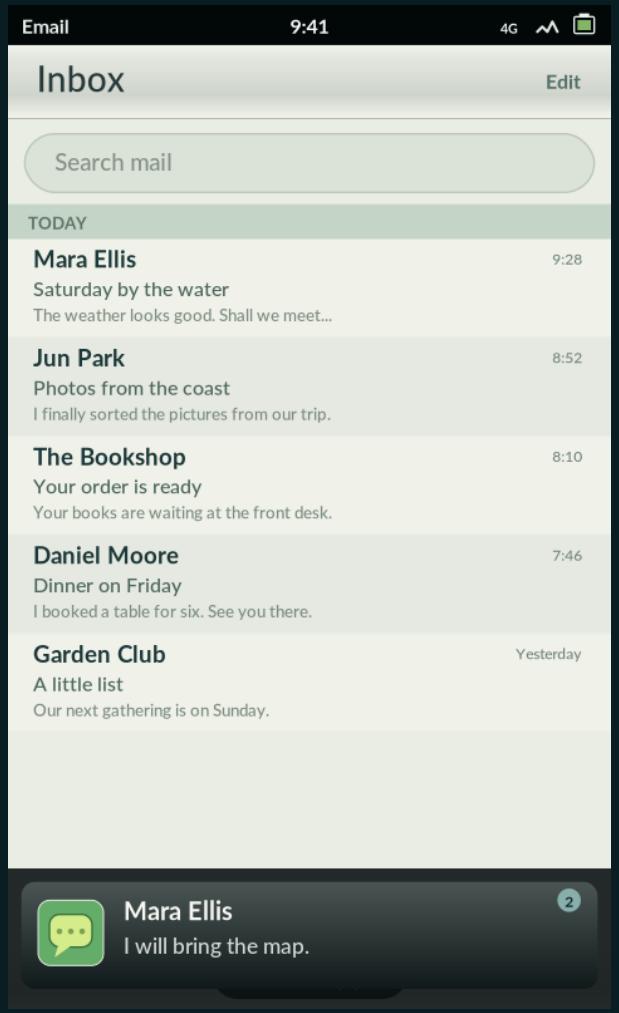


Email

Cooperating light application chrome

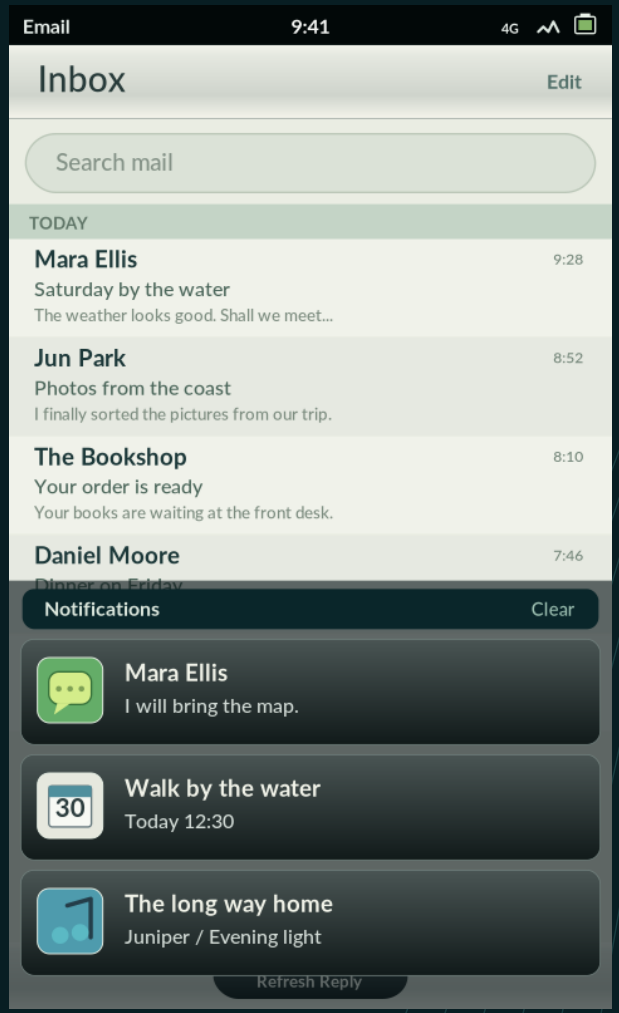
05–06 / screen studies

Editable SVGs preserve component geometry; PNGs verify native resolution.



Notification banner

Bottom arrival, foreground retained

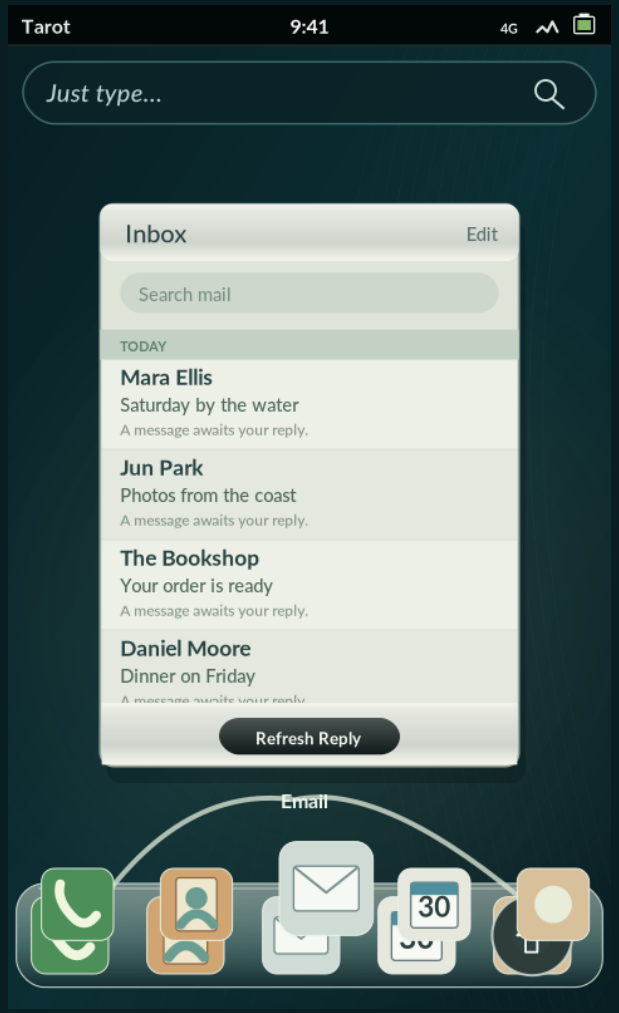


Dashboard

Persistent, actionable bottom history

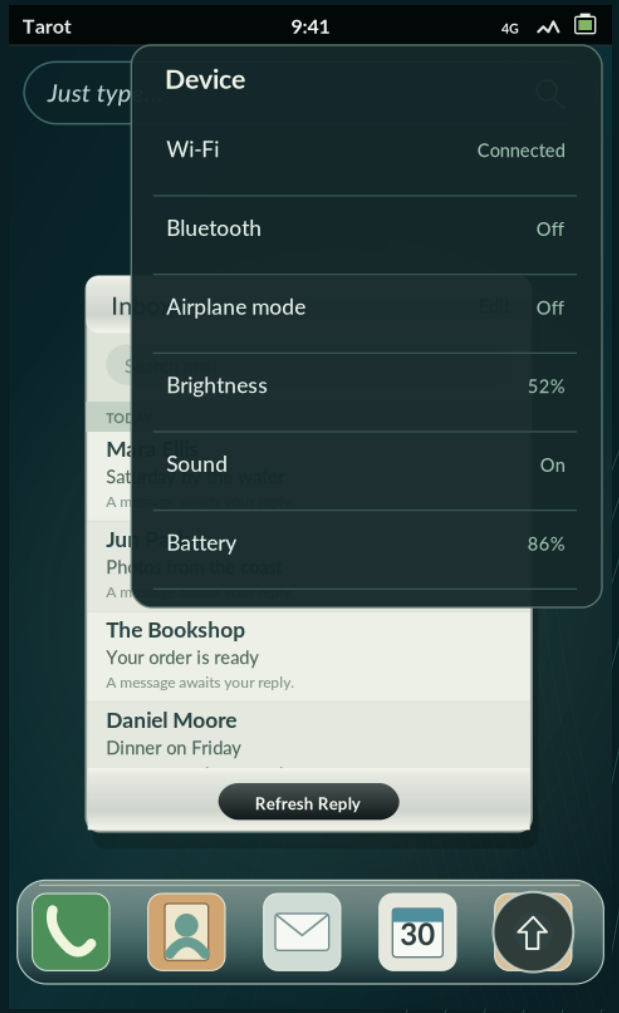
07–08 / screen studies

Editable SVGs preserve component geometry; PNGs verify native resolution.



Quick Launch wave

Hardware gesture area reveals favorites

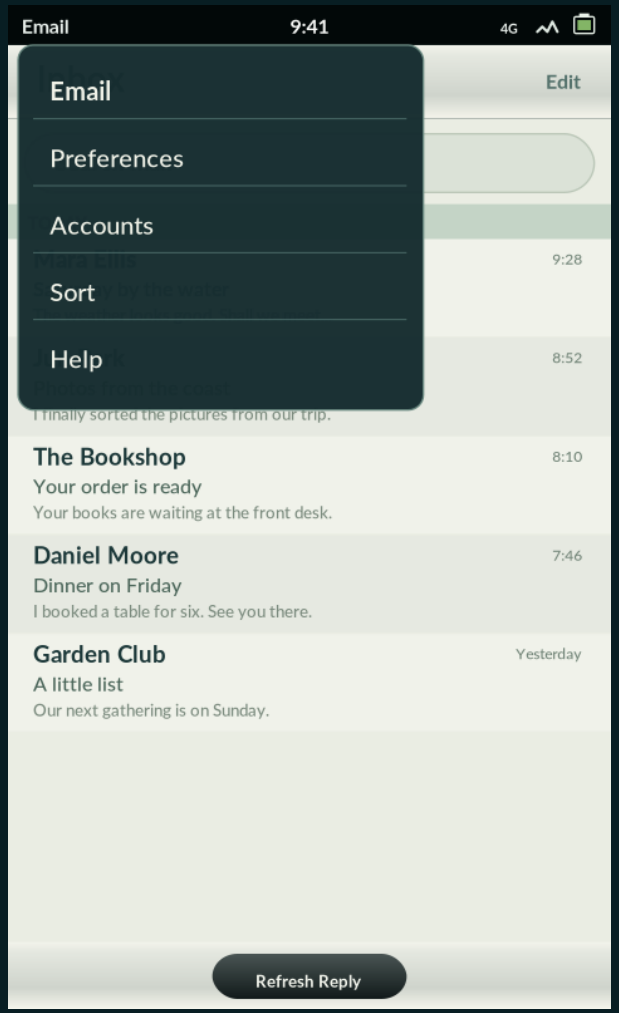


Device menu

Status corner, real services

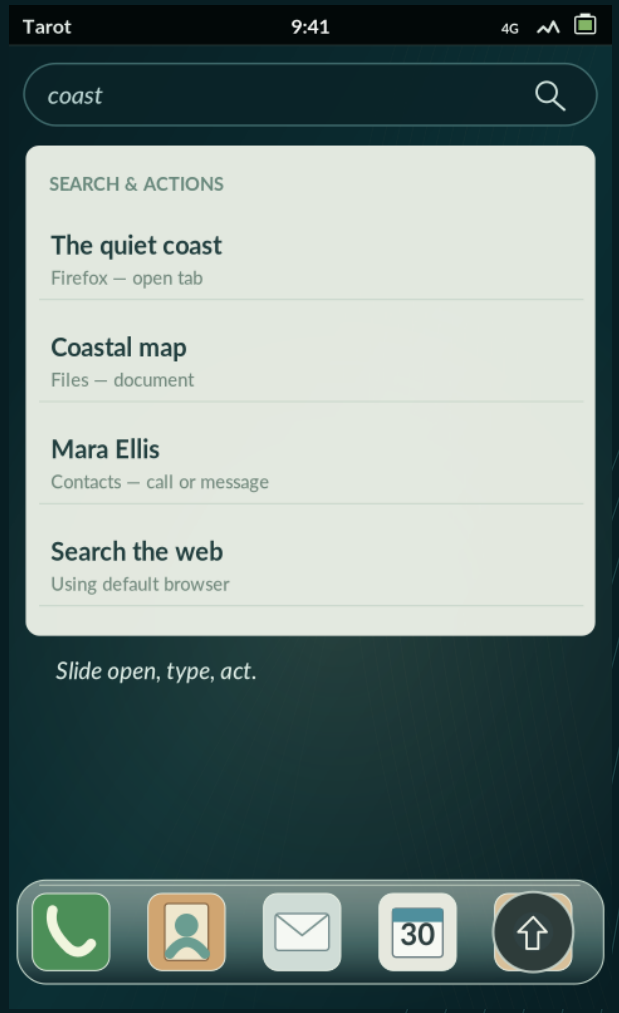
09–10 / screen studies

Editable SVGs preserve component geometry; PNGs verify native resolution.



Application menu

Top-left app actions

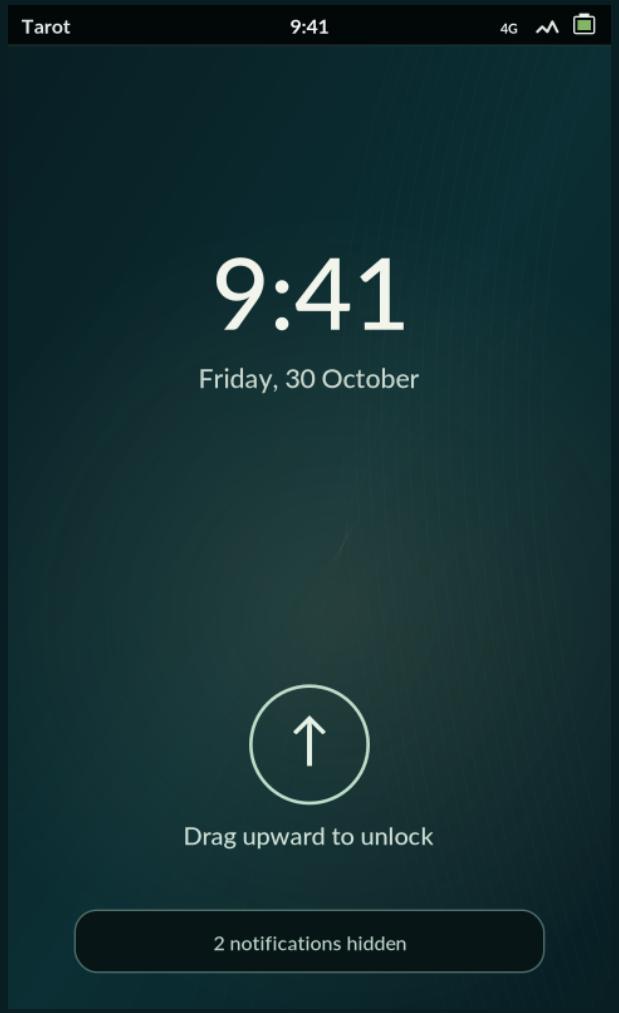


Just Type

Hardware keyboard starts intent

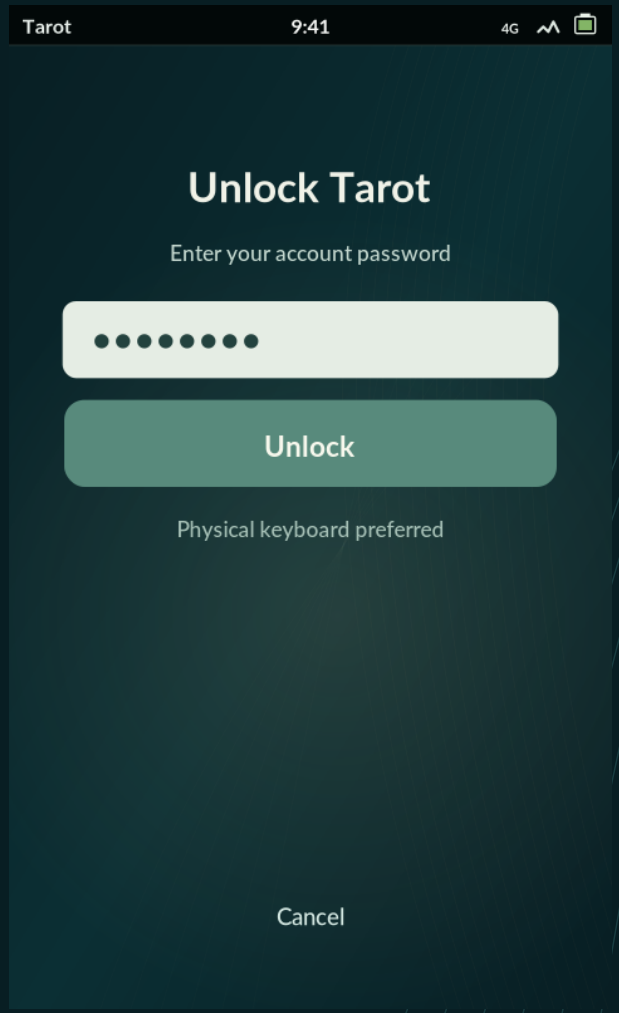
11–12 / screen studies

Editable SVGs preserve component geometry; PNGs verify native resolution.



Lock screen

Private notifications, protected compositor

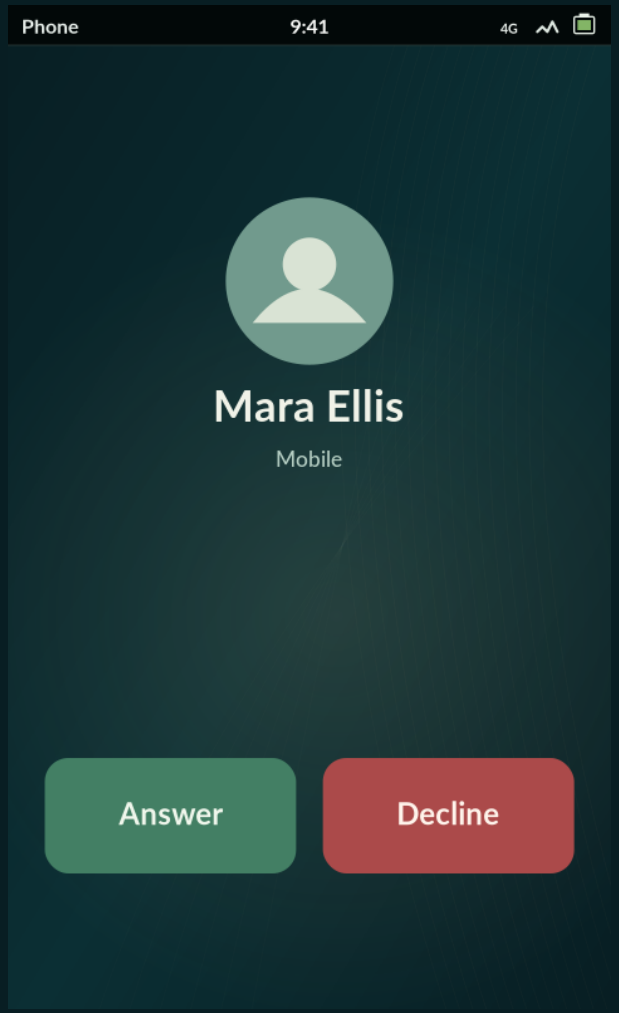


Authentication

Keyboard-first account credential

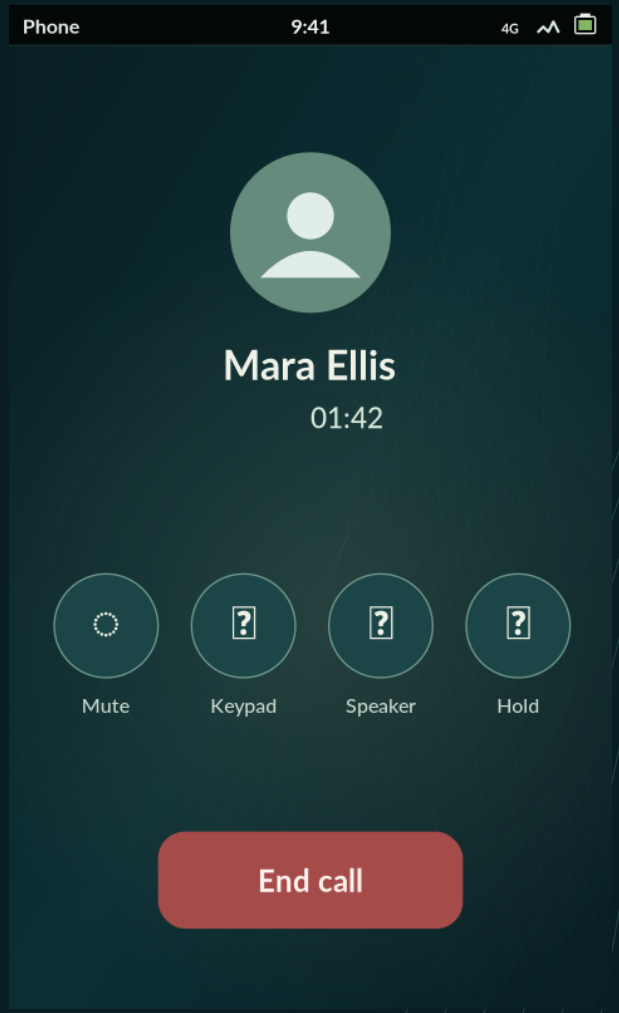
13–14 / screen studies

Editable SVGs preserve component geometry; PNGs verify native resolution.



Incoming call

Trusted urgent controls

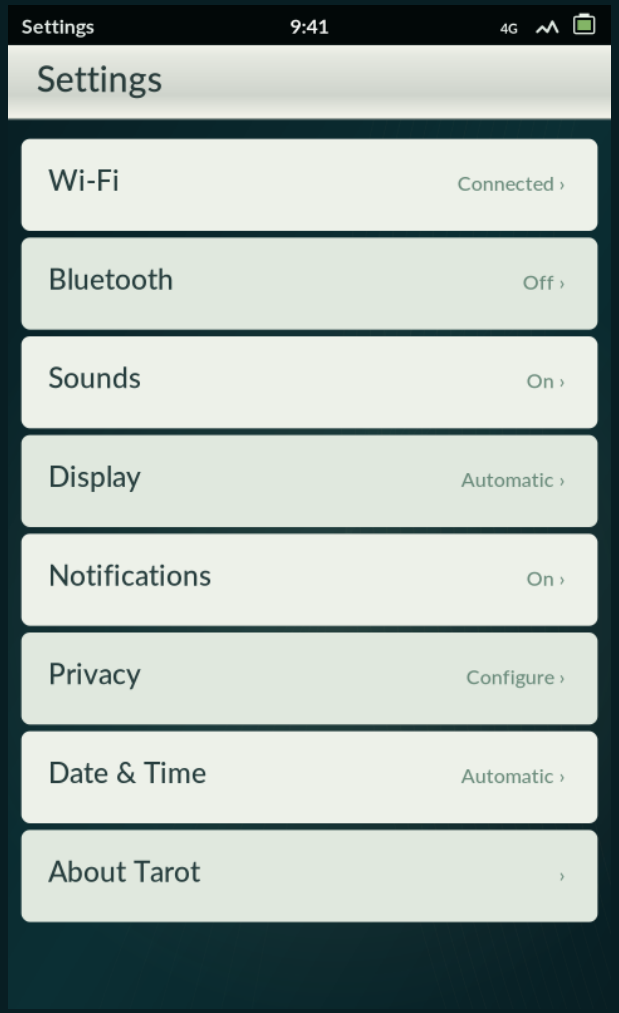


Active call

Call-state truth and routes

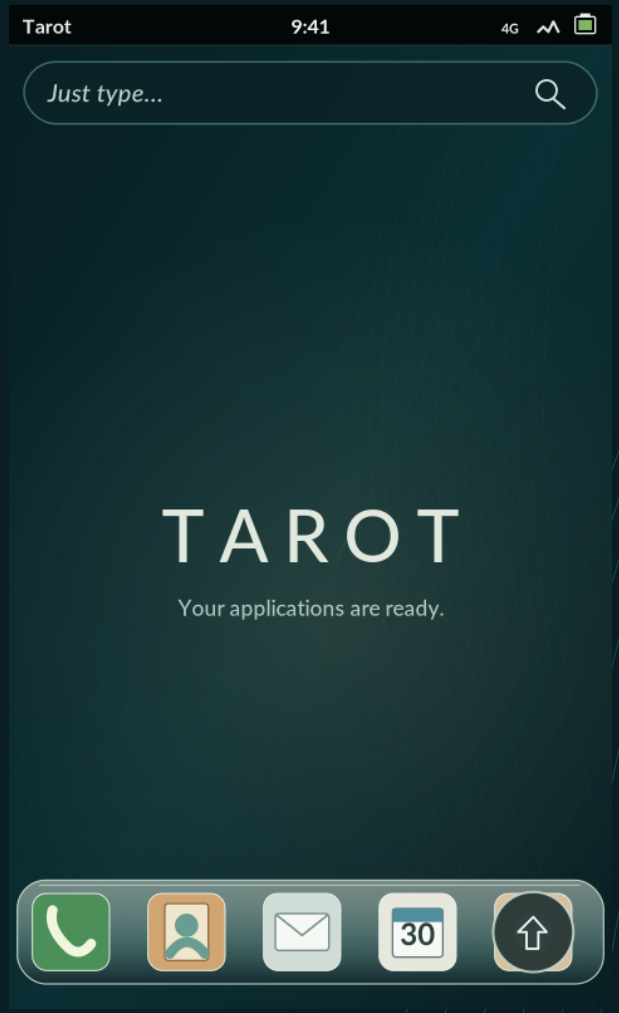
15–16 / screen studies

Editable SVGs preserve component geometry; PNGs verify native resolution.



Settings

Warm tactile list and toggles

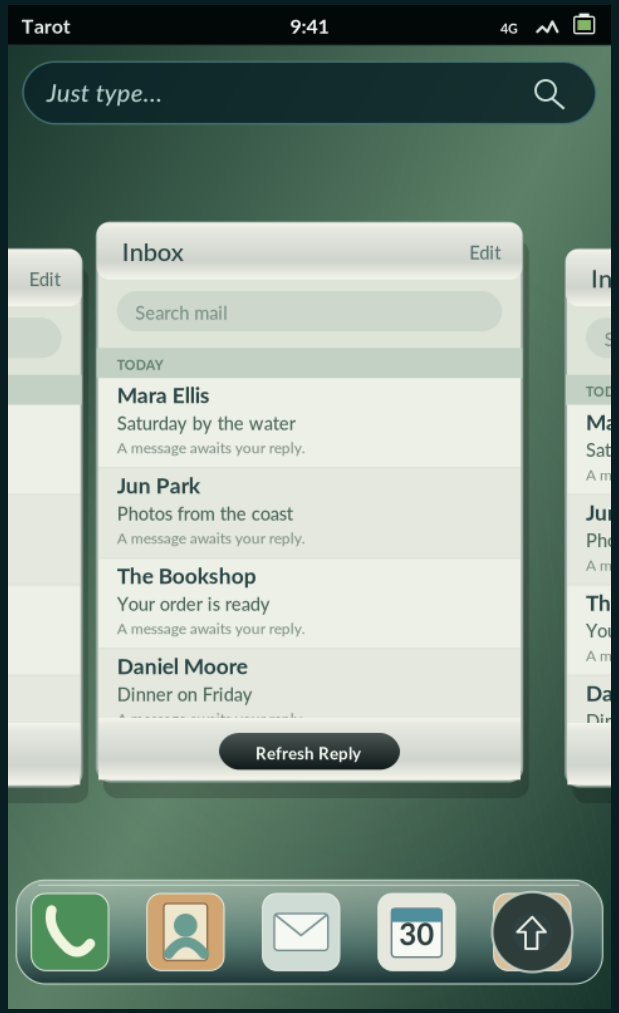


Empty card view

Wallpaper and dock carry composition

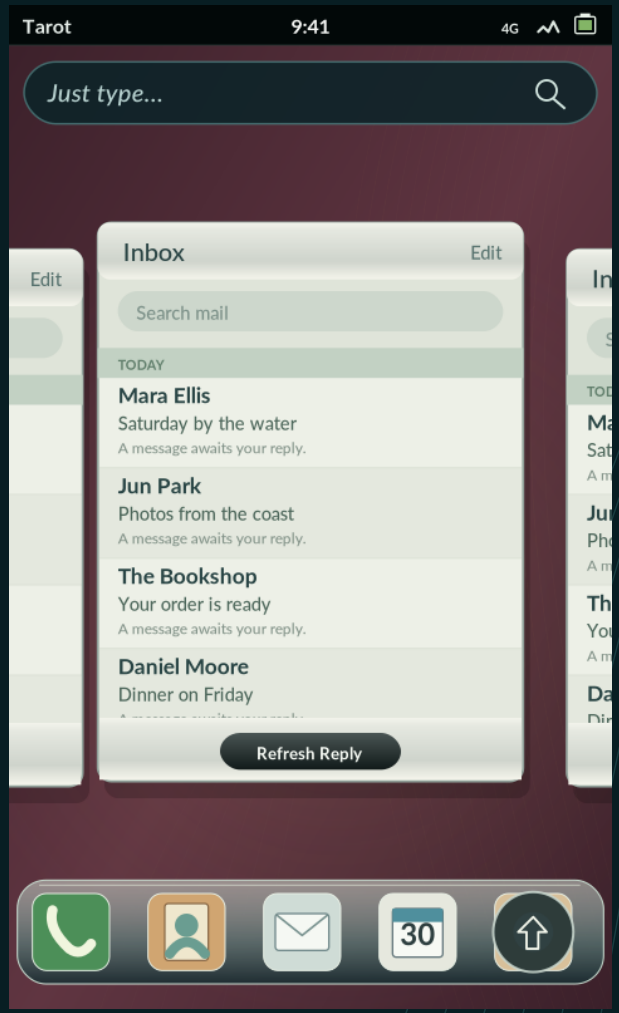
17–18 / screen studies

Editable SVGs preserve component geometry; PNGs verify native resolution.



Heritage

Period color alternative

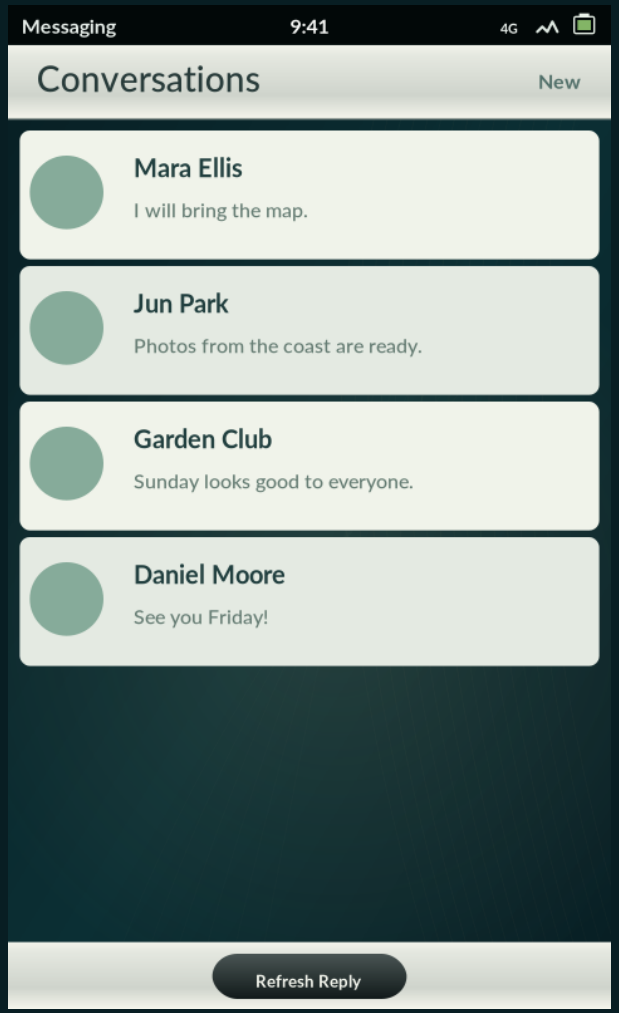


Vellum

Warm editorial alternative

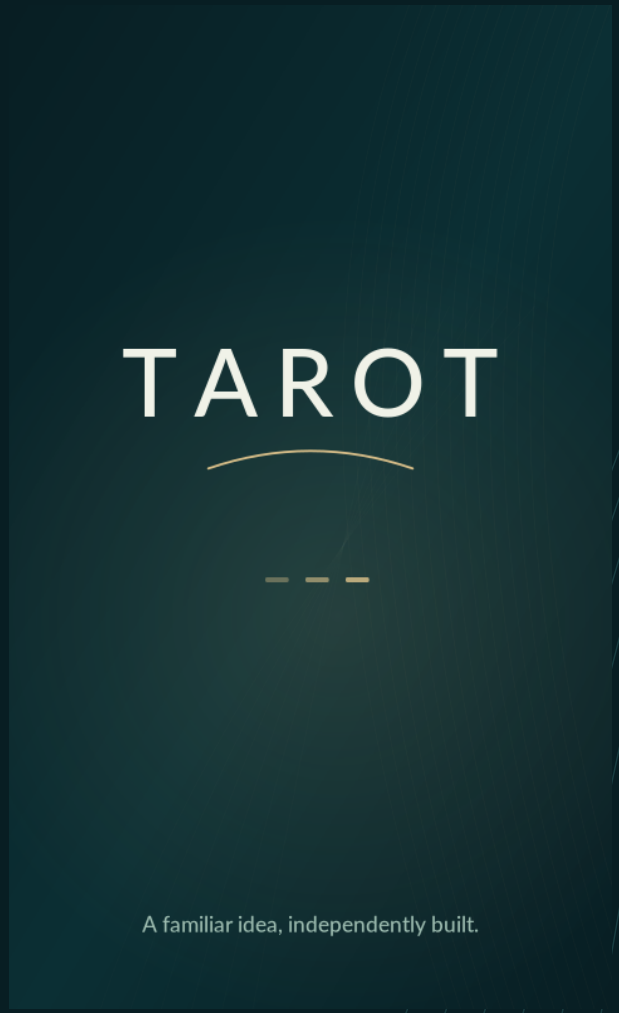
19–20 / screen studies

Editable SVGs preserve component geometry; PNGs verify native resolution.



Messaging

Conversations and input ownership

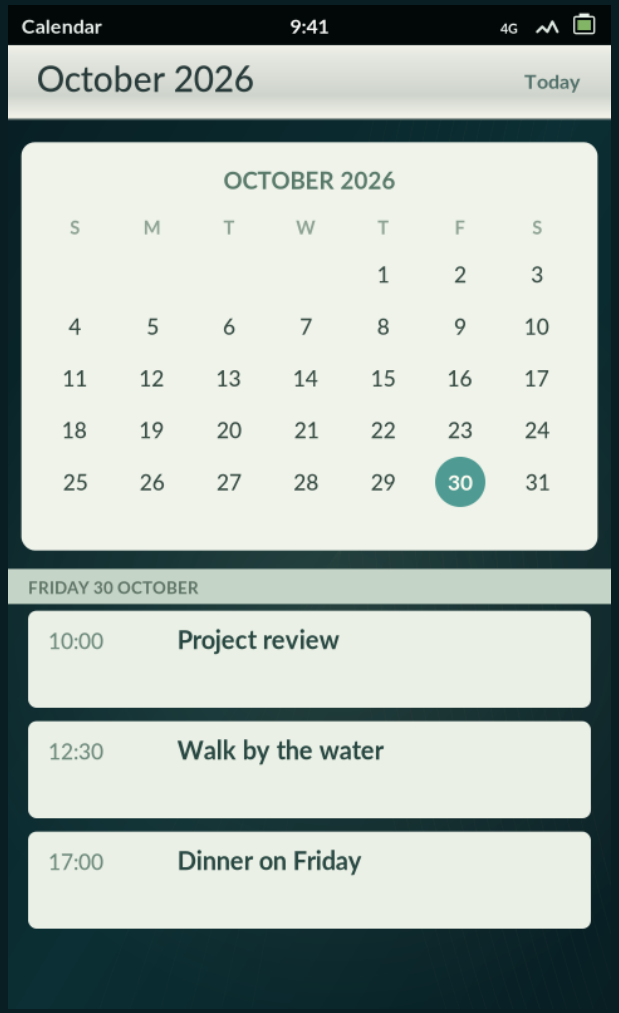


Boot

Quiet Tarot identity and startup

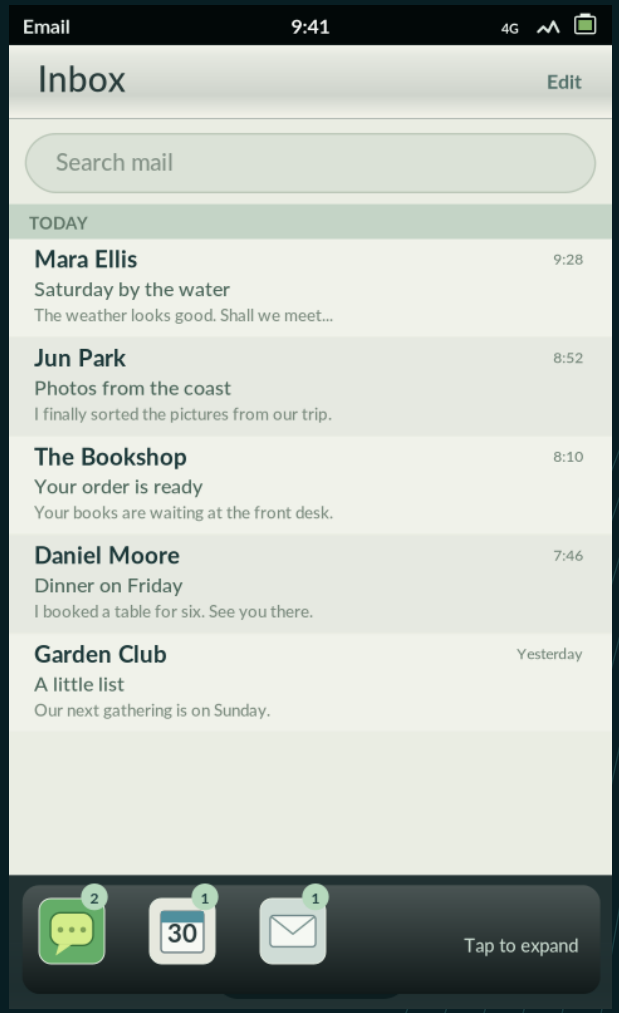
21–22 / screen studies

Editable SVGs preserve component geometry; PNGs verify native resolution.



Calendar

Date and schedule

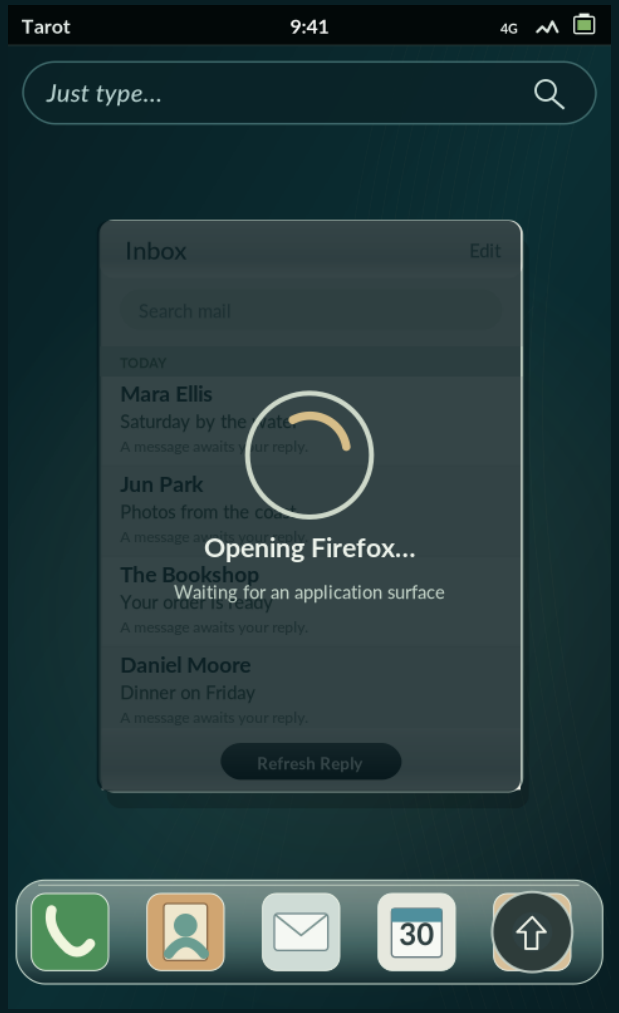


Notification rail

Persistent counts at the base

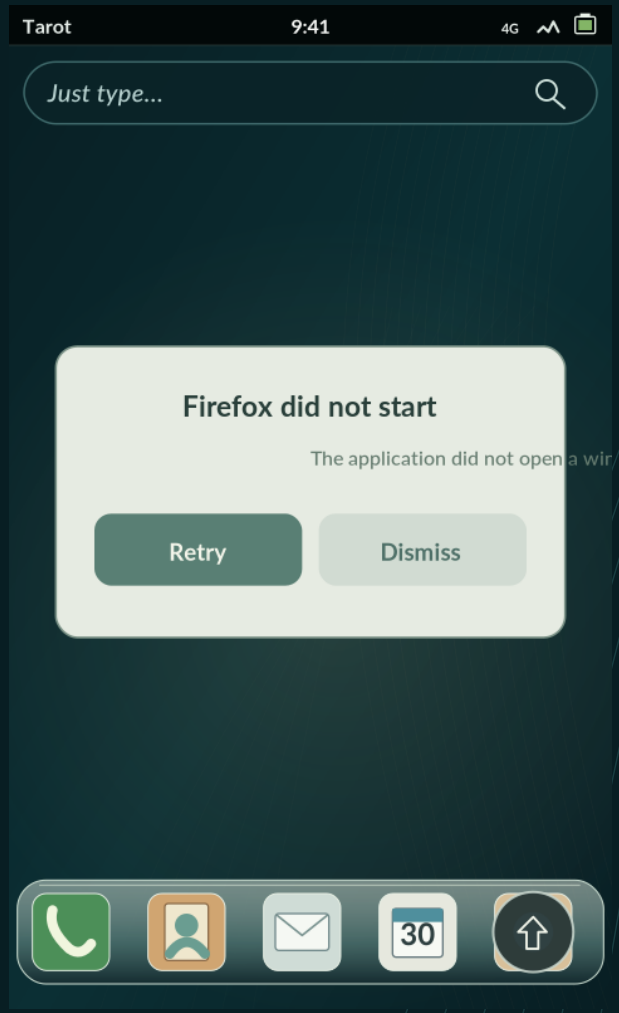
23–24 / screen studies

Editable SVGs preserve component geometry; PNGs verify native resolution.



Launch pending

One identity, honest waiting state

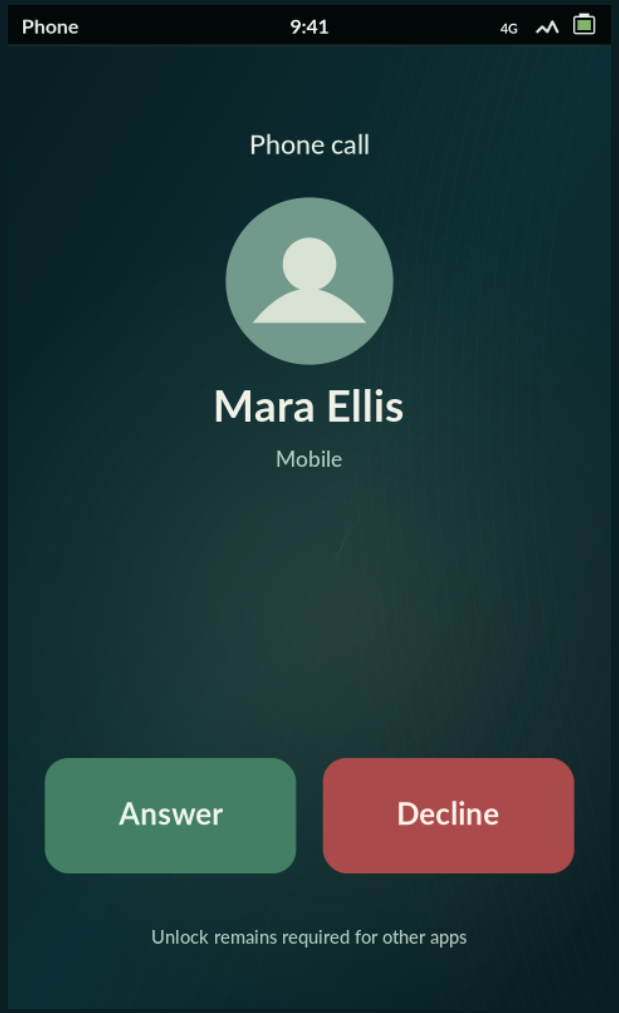


Launch failed

Recover without ghost windows

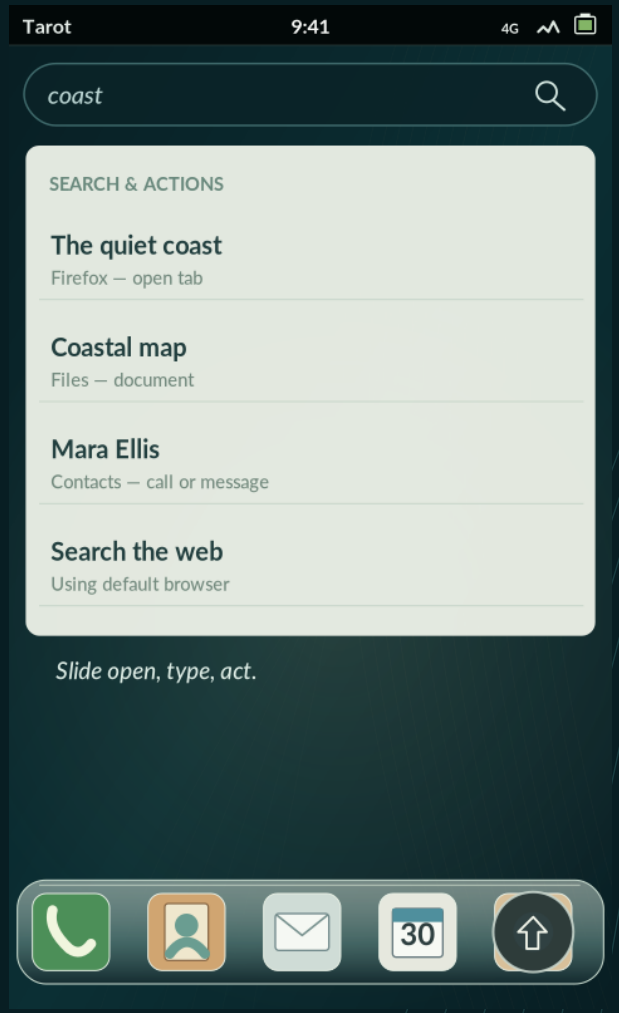
25–26 / screen studies

Editable SVGs preserve component geometry; PNGs verify native resolution.



Locked incoming call

Answer without leaking lock context

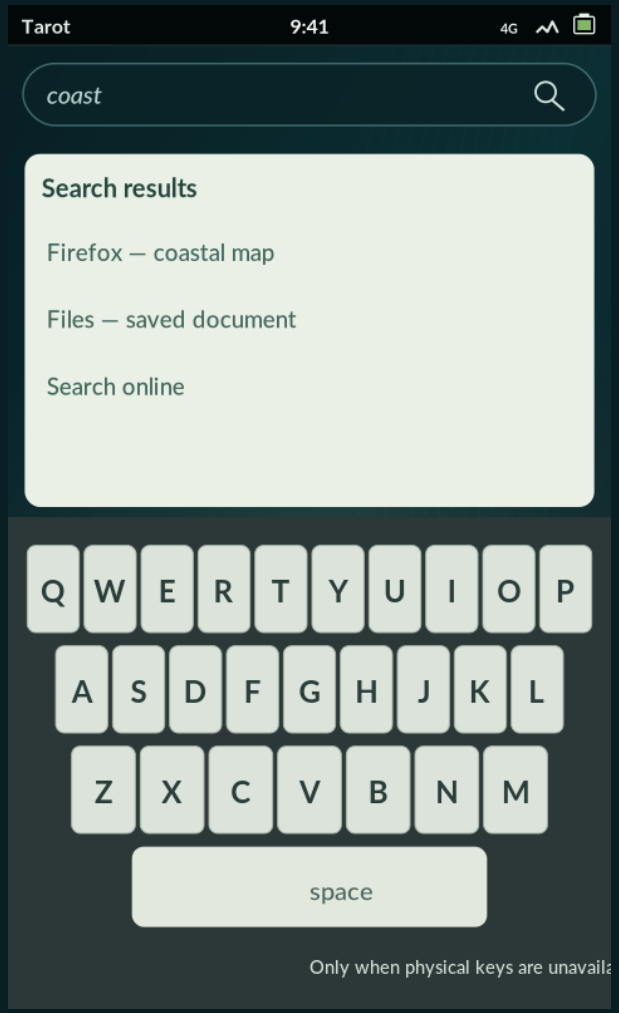


Slider open

Just Type begins with physical keys

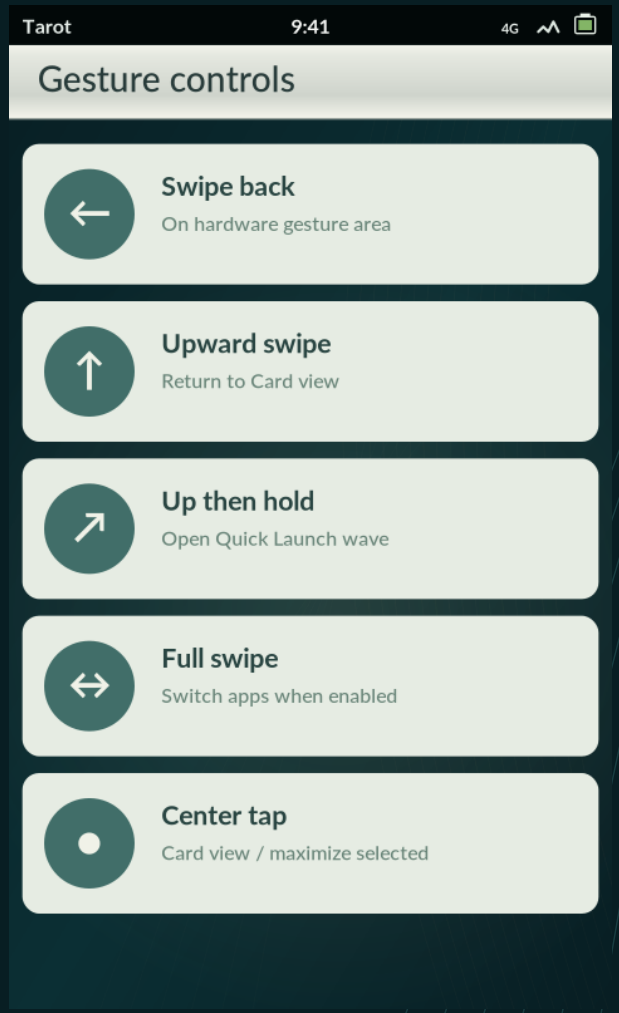
27–28 / screen studies

Editable SVGs preserve component geometry; PNGs verify native resolution.



Keyboard fallback

On-screen keys only where needed

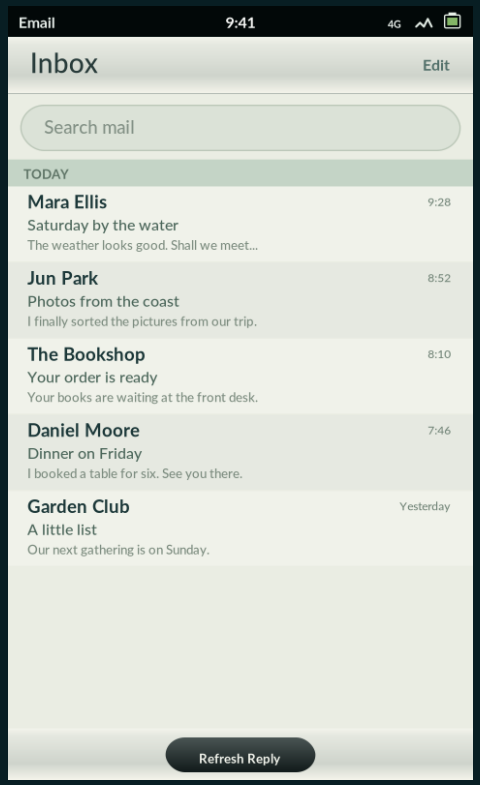


Gesture area guide

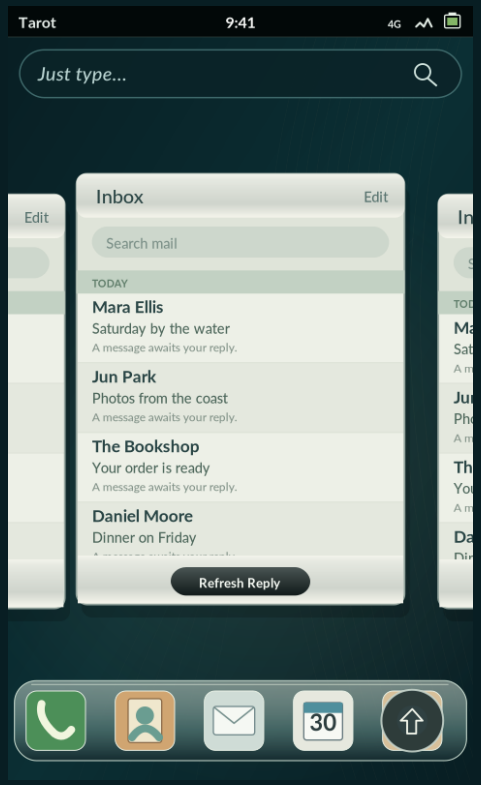
Hardware-origin gestures explained

Motion study

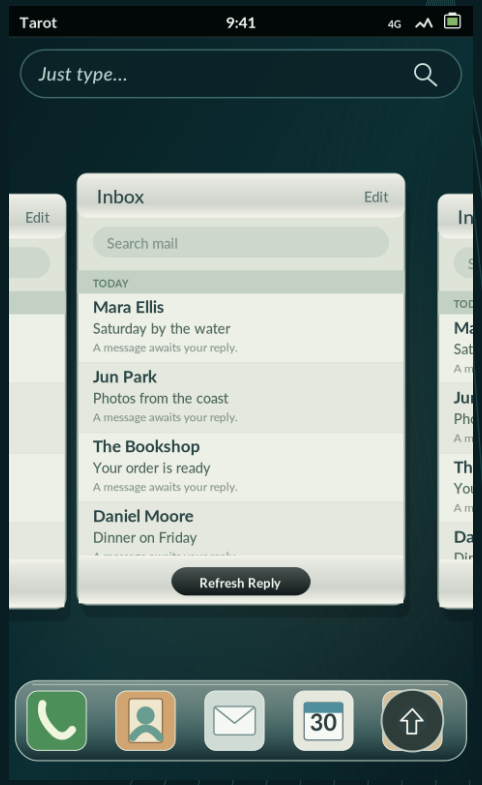
Card motion is proportional to a physical gesture, not a fixed screenshot transition.



Start: native window



Middle: live transform



End: card in overview

Reference seeds: minimize 350 ms / maximize 300 ms / dismiss 300 ms / launch 400 ms.

Trusted states and risk boundaries

No fidelity detail may undermine security, focus or accessibility.

Lock	Render a protected scene before accepting sensitive input. Never leak app thumbnails or notifications.
Auth	PAM is authoritative. Slide key and gesture hardware must not bypass the protected input owner.
Calls	Incoming calls use trusted urgent controls; call state comes from ModemManager, not animations.
Menus	Wayland client headerbars remain client-owned unless the client cooperates.
Close	Card flick requests protocol close. Confirmation dialogs remain visible and accessible.
Text entry	Physical key maps, repeated keys, dead keys and accessibility IME require a test matrix.

Port handoff: assumed Nura / postmarketOS

A design assumption is not a claim of verified device hardware support.

Display	480 × 800 output, scale 4/3; direct KMS mode, rotation, format and refresh to inspect.
Input	Capacitive gesture sensor Linux event shape, slider switch, keyboard keymap, keyboard illumination.
GPU	Adreno driver stack and available EGL/GLES/GBM; no assumed Panfrost or cross-GPU path.
Power	Backlight, proximity, suspend/resume, wake sources, thermal and charging policies.
Telephony	Radio/modem compatibility, voice routing, call notification, emergency-call safety tests.
Packages	Tarot as compositor session, protected login, native app launcher, constrained RAM behavior.

Fidelity acceptance

A screen-by-screen comparison is needed before calling the UI indistinguishable.

A Historical build & locale: record exact Pre 3 variant, firmware, timezone and captured state.

B Status and icons: compare baseline, menu-open, focused, overview, lock and radio-unknown states.

C Card continuity: capture one app across launch, minimize, stack, reorder, dismiss and refusal.

D Gestures: verify physical sensor origin, trajectory, cancellation, tap/back and wave arbitration.

E Slider: open/close focus, live keymaps, Sym/Option, search, clipboard and safe wake.

F Native clients: GTK/Qt/Firefox/Thunderbird parent dialogs, popups, text input and fullscreen.

G Resource limits: cold/warm app startup, compositor RSS, GPU memory, 20+ sleep cycles.

H Accessible fallback: enlarged text, gesture alternatives, reduced motion, physical key failure.

Research and licensing

Original vectors are studies; historical documentation remains external.

Pre 3 manual / hardware

hp.manymanuals.com/pc-workstation-barebones/pre3/users-guide-1137/12

Pre 3 manual / Card view

hp.manymanuals.com/pc-workstation-barebones/pre3/users-guide-1137/13

Pre 3 manual / stacks

hp.manymanuals.com/pc-workstation-barebones/pre3/users-guide-1137/43

Pre 3 manual / Just Type

hp.manymanuals.com/pc-workstation-barebones/pre3/users-guide-1137/27

HP Pre 3 hardware specifications

gsmarena.com/hp_pre_3-3770.php

Tarot implementation / current architecture

github.com/timschmidt/tarot

No historical manual images, proprietary Palm fonts or extracted firmware assets are distributed.

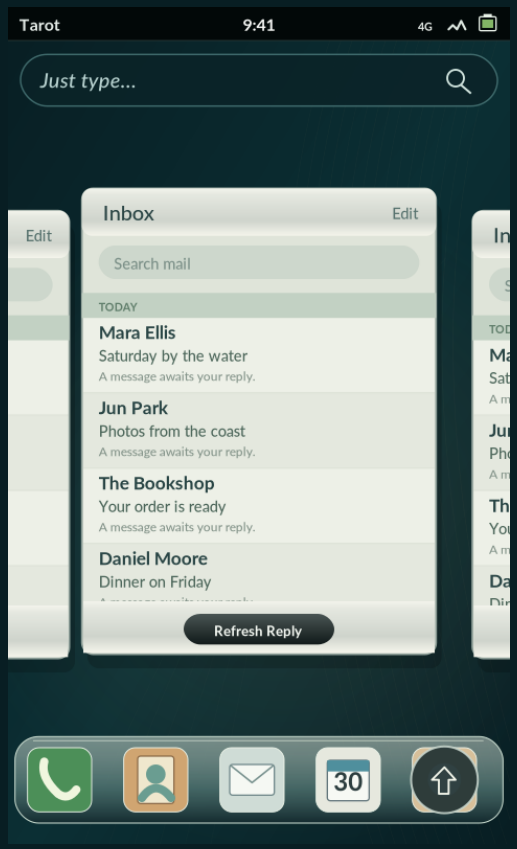
Implementation sequence

Respect the existing PinePhone Pro work and add Pre 3 via a device profile.

01 Device profile	Introduce native 480 × 800 / 360 × 600 and input capability flags.
02 Hardware test	Prove keyboard, slider, gesture sensor, KMS, authentication and suspend.
03 Shell layout	Render live Card view, Launcher, status, Quick Launch and dashboard at native size.
04 Interaction test	Close/refuse, stack, gesture wave, Just Type from physical keys.
05 Visual QA	Use 28-state atlas and same Pre 3 firmware build reference; record deviations.
06 Optional bridges	Firefox/Thunderbird semantic cards and Waydroid only after resource validation.

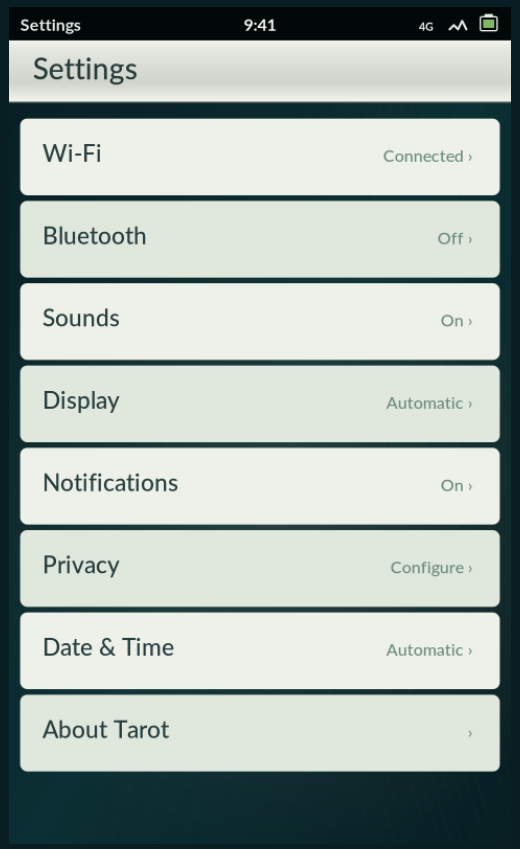
One metaphor, three devices

Tarot can be faithful without pretending hardware differences do not exist.



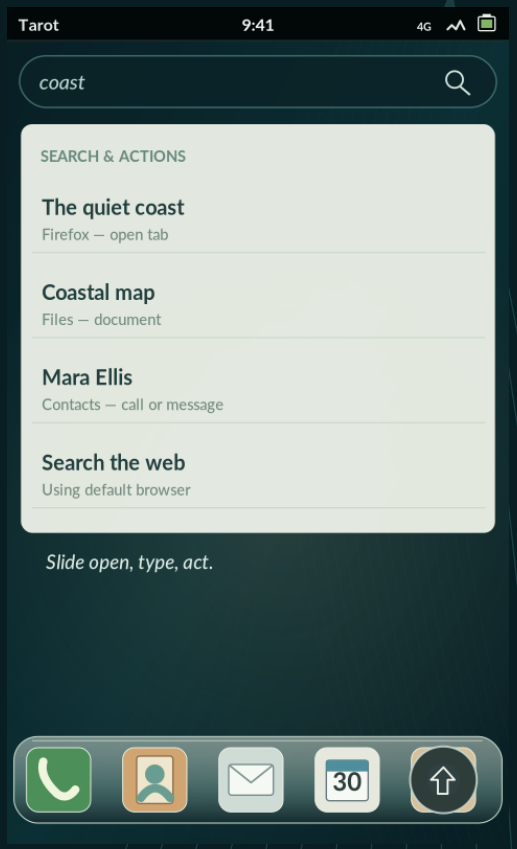
PinePhone Pro

Soft gesture strip



LG Fx0

Gold theme + Home key



HP Pre 3

Hardware gesture sensor + slider

Shared card model and design language; the Pre 3 edition is a standalone companion, not a replacement.